

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		add soap solution to each sample and shake (1) the sample that produces a lather is soft water (1) boil the remaining samples (1) the sample that now gives a lather is temporary hard water and the one that still does not give a lather is permanent hard water (1)	2		2	4		4
		(ii)		Z is temporary hard water / has significant temporary hardness with a small amount of permanent hardness (1) it has a high concentration of hydrogencarbonate ions / it has a high concentration of hydrogencarbonate ions and a small concentration of sulfate ions (1) ignore reference to high magnesium ion concentration			2	2		

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					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		hard water contains magnesium ions / calcium ions / Mg^{2+} / Ca^{2+} (1) swap places with two sodium ions (1) accept replace / exchange for swap must be one reference to ions for full credit	2			2		2
		(ii)		either of following - all sodium ions have been used up - no more sodium ions left	1			1		1
	(c)			$0.00135 / 1.35 \times 10^{-3}$ (2) if answer incorrect award (1) for $M_r \text{CaSO}_4 = 136$ award (1) only if answer not given to three significant figures ecf possible from incorrect M_r		2		2	2	
				Question 7 total	5	2	4	11	2	7